

Title: Cartographic Mapping of Natura 2000 Protected Areas and Anthropogenic Activity in the Thermaic Gulf

This cartographic project was implemented within the **QGIS** software environment to visualize and analyze the spatial interaction between the protected ecosystems of the **Natura 2000** network and human activity (urban fabric, road networks) in the broader region of Thessaloniki.

To develop the map, point and line geospatial data (buildings, roads) were combined with a high-resolution satellite background. The Natura 2000 zones (Axios-Aliakmonas Delta, Angelochori Lagoon, Epanomi Tsairi, and Kitros Lagoon) were digitally drafted using the vector polygons creation method. The geometry and coordinates of their boundaries were based on official digital data and spatial specifications acquired from local government open databases.



The Significance and Utility of Delineating Natura 2000 Areas

The digital delineation of Natura 2000 areas through **Geographic Information Systems (GIS)** is not merely a drafting process, but a crucial tool for environmental protection and sustainable development. Accurate mapping of their boundaries serves three core pillars:

- **Institutional Protection and Legal Shielding:** Digital boundaries define with absolute precision the zones where specific human activities are prohibited or restricted (e.g., industrial construction, intensive agriculture, unauthorized tourism development). They form the legal foundation for enforcing environmental legislation.

- **Proper Spatial Planning:** It enables urban planners, engineers, and local authorities to know in advance where residential/commercial zones end and protected nature begins. In this way, destructive interventions in sensitive ecosystems (such as the wetlands of the Thermaic Gulf) are avoided.
 - **Biodiversity Management and Protection:** Polygon mapping helps scientists monitor ecosystem connectivity, establish "green corridors" for wildlife migration, and design targeted conservation measures for endangered species and their habitats.
- Overall, this map serves as a valuable decision-making tool, visually demonstrating how human activity can and must coexist in balance with the natural environment.